

DrishtiLM

Legal Metrology Packaged Commodities Compliance Scanner

SMART INDIA HACKATHON 2026
Problem Statement ID: SIH26034

Prototype Demonstration Document
(Backup Showcase for Jury Evaluation)

Item	Detail
Team	_____
Institution	_____
Date	September 2026
Demo URL (local)	App http://localhost:3000 API http://localhost:8000/docs
Runs on	Ordinary laptop — CPU only, works offline

1 The Problem

Under the Legal Metrology (Packaged Commodities) Rules, 2011, every packaged product sold in India must carry mandatory declarations — manufacturer details, generic name, net quantity, MRP inclusive of all taxes, dates, customer-care contact and country of origin — printed in legally prescribed minimum sizes (Rule 7).

- Officers inspect packs **manually**: slow (about 15 minutes per pack), inconsistent between officers.
- **Print-size violations are invisible** in the field — nobody carries a microscope to check 1 mm lettering.
- **No evidence trail** — findings rest on memory and paper notes, weak in enforcement proceedings.
- Packers have **no self-check tool** before dispatching lakhs of labels to print.

What is needed: a mobile-first tool that photographs a pack, reads the label, checks every legal rule automatically, and produces an auditable, officer-signed report — on ordinary hardware, with no internet or GPU.

2 Our Solution

DrishtiLM is a field compliance scanner. An officer photographs a product package with a phone. The system cleans the image, reads all label text, extracts the legal declarations, **measures the physical print size in millimetres**, checks each declaration against the 2011 Rules, and gives a verdict — **COMPLIANT**, **NON_COMPLIANT**, or **INCOMPLETE** — with photo evidence and rule-wise reasoning. A human officer always confirms the result before it becomes final.

Why it is different

- **Measures print size physically** (Rule 7 font metrology from a phone photo) — photo-only tools cannot do this.
- **Deterministic, cited verdicts** — every check traces to the exact sub-rule and Gazette page, not an AI guess.
- **Honest uncertainty** — a weak or unmeasurable read returns **INCOMPLETE** (*take a better photo*), never a false guilty verdict or a silent pass.
- **Legal-grade human gate** — officer confirm / correct / override with full audit trail before any report is final.
- **Runs anywhere** — ordinary laptop or phone-server, CPU only, offline, one-command setup.

3 How It Works (8 steps)

- **1. Take photo** — officer photographs the pack (full view + close-up macro, reference card in frame).
- **2. Guided capture** — the app checks angle, sharpness and readability live; asks for a retake if blurry.
- **3. Secure upload** — signed-in officers only; photo, GPS and pack inputs reach the checking service.
- **4. Clean & measure** — image straightened and sharpened; text height measured in real millimetres via the card scale.
- **5. Read label** — every printed word read automatically, each with a confidence score.
- **6. Key details** — price, weight, dates, maker, helpline pulled out of the text.
- **7. Legal check** — each declaration tested against Rules 6 and 7 with exact citations.
- **8. Officer review & report** — officer confirms; signed PDF with proof photos unlocks.



DrishtiLM – How It Works

A simple overview for everyone · No technical knowledge needed

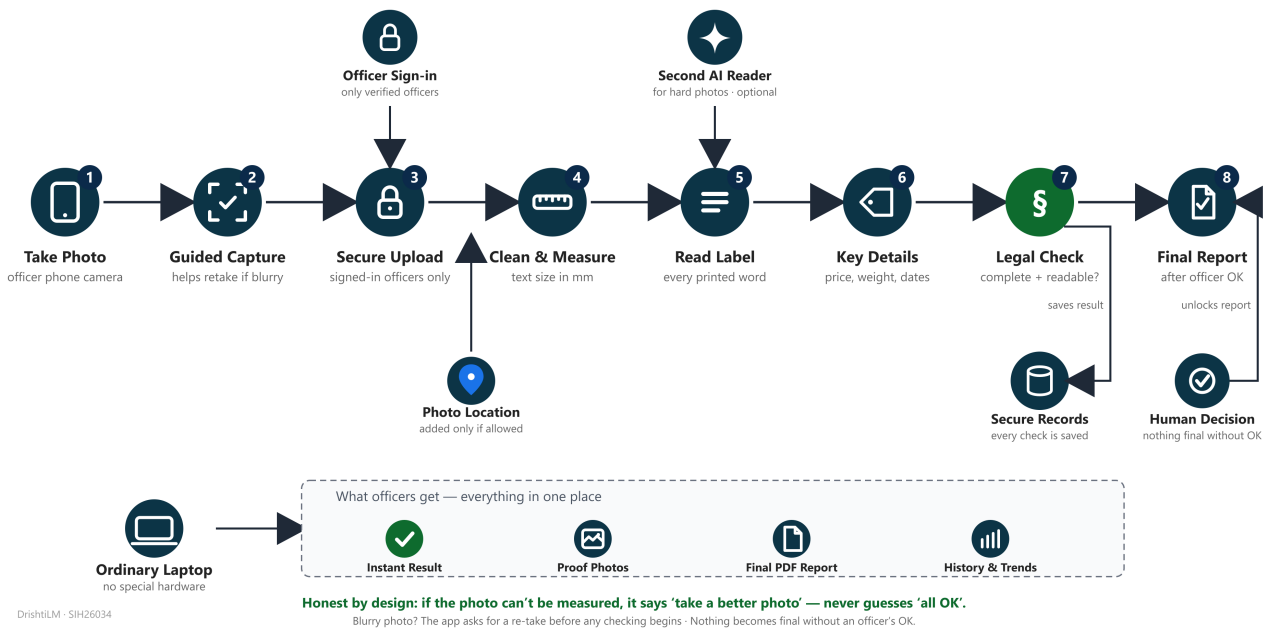


Figure 1 — Plain-language system overview: capture to signed report.

4 Live Demo Script (5 minutes)

If the live demo cannot run, this section plus Figure 1 is the backup showcase — it describes exactly what the jury would see.

Step	Action	Jury sees
1	Officer signs in, opens New Scan	Guided camera with alignment stencil
2	Photograph a compliant pack + a violating pack	Live readability check; retake prompt on blur
3	Submit; system reads and checks (~seconds)	Verdict banner: COMPLIANT vs NON_COMPLIANT with rule cards
4	Open a failing rule	Why it failed + exact next steps (retake / verify / correct)
5	Show a blurry-photo scan	INCOMPLETE — 'take a better photo', never condemned
6	Review: confirm one scan, correct one field	Audit trail (who, when); verdict refolds
7	Download PDF	Signed report: findings table, proof photos, enforcement summary
8	Dashboard	History, statistics, top-violated rules

5 Legal Rule Coverage

Rule	What the system checks
6(1)(a)	Manufacturer / packer / importer name + address present
6(1)(b)	Generic / common commodity name present
6(1)(c)	Net quantity present, in standard units
6(1)(e) + 2(m)	MRP present, greater than zero, with 'inclusive of all taxes'
6(1)(d)	Manufacture month/year present; expiry after manufacture

Care proviso *	Customer-care name / address / contact present
Origin *	Country of origin declared for imported packs
7(2) Table-I/II	Numeral height vs tier (1 / 2 / 4 / 6 mm; doubled if embossed)
7(3)	Letters at least 1 mm (2 mm embossed); width at least one-third of height

* Post-2011 amendment wording — flagged honestly as unverified in the tool; all other citations verified against the official Gazette PDF stored with the project.

6 Technology (CPU-only, offline)

- **App:** Next.js + React (mobile-friendly) talking to a FastAPI backend over a versioned REST API.
- **Vision/reading:** OpenCV cleanup + reference-card millimetre scale; Tesseract OCR with per-word confidence.
- **Understanding:** pattern extraction + trained language model for declarations, maker-name directory, barcode cross-check.
- **Brain:** pure deterministic rule engine — no black box decides guilt.
- **Records:** PostgreSQL (production) / SQLite (local); signed PDF reports; role-based logins; audit trail on every human action.
- **Quality:** 120+ automated tests, CI on every change, one-command Docker deployment.

7 Prototype Status

- Working end-to-end today: capture, read, extract, check, review, PDF, dashboard.
- Tested on synthetic labels plus real market pack photos; officer corrections feed back into model retraining.
- Known hard cases (embossed date stamps, glossy curves) are handled by guided retake — the system says so openly.

Roadmap

- Regional-language label reading.
- Video live-scan mode and department case-system integration.
- Accuracy leaderboard per field, published with every model update.

DrishtiLM turns a 15-minute manual label inspection into a 1-minute guided scan with photo evidence, cited legal reasoning and an officer-signed report — accurate where it can read, honest where it cannot.